

**In Support of Federal Funding for the
National Artificial Intelligence Research Resource (NAIRR)**

**San Diego Supercomputer Center
(UC San Diego)**

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The United States of America presently dominates AI globally because of our intellectual and technological leadership in hardware, software, and algorithms. To sustain and enhance this dominance requires a robust ecosystem that includes—in addition to industry across all sectors—a thriving start-up culture, our national laboratories, and our world-leading academic research and education enterprise. Higher education not only trains the workforce of the future; it makes the discoveries of the future and translates foundational discoveries into practical applications for science, industry, and society.

We recommend that the federal government turn the National AI Research Resource (NAIRR) Pilot program into a fully fledged and funded program that leverages the complementary strengths of federal agencies, national laboratories, private foundations, the national university research enterprise, and industry—from small start-ups to the world’s largest companies. University research centers have been critical to the implementation of the NAIRR Pilot and are strategically placed for leadership in a full NAIRR program.

In the following, we explain why university research centers should be an essential part of the AI Action Plan via examples of some of the San Diego Supercomputer Center’s activities that are exemplars for such research centers throughout the country. Centers like SDSC are national assets in maximizing the return on federal investments in AI and ensuring that the U.S. remains at the cutting edge.

The San Diego Supercomputer Center, part of the School of Computing, Information, and Data Sciences at UC San Diego, is uniquely positioned to complement the multi-sector strengths that comprise the nation’s thriving AI ecosystem. Founded in the 1980s to support the nation’s growing scientific computing and data processing needs, SDSC is one of the premier academic centers for high-performance and data-intensive

computing. Over the last four decades, SDSC has enabled and led groundbreaking advancements in data and computationally intensive science.

Today the center harnesses its long-standing expertise to advance AI innovation. SDSC strategically bridges gaps needed for discovery within the multi-sector AI landscape. This notably includes translating data and fundamental science into actionable knowledge and products; catalyzing AI-powered breakthroughs to accelerate discovery in high-impact fields—from fusion energy to personalized medicine to disaster management and prevention—enabling the incubation of discoveries that can later be brought to scale through industry partnerships; providing curated solutions to academic and industry researchers through open-source software development and maintenance; and collaborating with industry to make the use of data and compute resources needed for AI-intensive scientific research in academia possible. **Maximizing the continued impact that institutions like SDSC can have for the nation necessitates strategic federal funding investments in research on AI algorithms and systems, as well as the full range of applications of AI to science, technology, and industry.**

We provide two specific examples of long-standing programs led by SDSC that demonstrate how federal investments have had a role in catalyzing the advancement of cutting-edge scientific discovery that positively impacts the nation. These cases are models for research centers across the country, with different foci, that are developing and applying AI advancements to address pressing challenges facing the nation. These institutions nationwide are collectively an invaluable resource that the AI Action Plan can and should tap into via the creation of NAIRR.

For over a decade, SDSC has developed a research portfolio that applies AI to improve outcomes related to wildfires through cross-sector partnerships and collaboration. Research outcomes include mountaintop cameras, backcountry communications networks, multi-modal data commons, fire detection, and modeling algorithms, to name just a few. Research outcomes are operationalized to inform wildfire management and firefighter training to both combat active fires and mitigate the risk of future fires. This includes modeling the spread of active wildfires to protect public safety and property in real time. Meritoriously awarded federal research funding had an important role in the development of these efforts, which have gone from research incubation to fully-fledged tools and services scaled through public-private-research partnerships with utilities, state agencies, and the private sector. This research saves lives and reduces the economic devastation caused by wildfires.

The second highlighted research program collects and analyzes internet data in support of protecting national security. This portfolio developed technology that uses deep

learning that allows the U.S. military to use commercial 5G networks while minimizing risk to adversarial control or observation of communications. Funded initially by the federal government, this work resulted in the founding of a start-up company to commercialize some of the technology. Follow-on research efforts have focused on better understanding the global Internet through AI-driven insights in support of informing national security and engineering advancements.

In addition to leading projects that address profound domestic challenges, SDSC and university research centers have had a pivotal leadership role in enabling AI-intensive discovery—including through the NAIRR Pilot.

The NAIRR program, a result of the National AI Initiative created during the first Trump Administration, is unparalleled in its scope across more than a dozen federal agencies, industry, private foundations, and academia. Its blueprint envisions compute, data, software, and training resources in AI for the nation. An initial pilot program was started in early 2024 by NSF, DOE, NIH, NOAA, and NASA under the leadership of the NSF. It includes resources from DoD, DOE, NASA, NIH, NOAA, USDA, USGS, and USPTO, as well as national HPC systems run by university research centers like SDSC. Industry partners (AWS, Azure, Google, NVIDIA, OpenAI, Anthropic, Hugging Face, Groq, Cerebras, SambaNova, Databricks, Weights & Biases, and others) have contributed access to some of their products and services at substantial levels.

SDSC is presently making many contributions to this NAIRR Pilot. The SDSC CloudBank project manages the several tens of millions of dollars in cloud credit committed by private industry, including AWS, Azure, and Google. As a next step, CloudBank is working with the SDSC Sherlock program to allow for the use of regulated data via these cloud credits. Sherlock provides data management and access solutions for regulated data. We provide scientific research computing support to select science projects on the NVIDIA DGX Cloud that enabled the consumption of \$30M in cloud credit, so far, committed by NVIDIA to the NAIRR Pilot. These commercial resources all complement the national HPC systems, run by university research centers like SDSC, that are available through the NAIRR Pilot. SDSC develops and operates immersive digital environments for science, called “Science Gateways,” that have more than 2M unique users per year. SDSC has a wide range of projects focused around the translation of data into actionable knowledge, arguably one of the most compelling needs of modern times. Those projects cover the entire vertical stack from storage to content delivery networks to the meaning of data and knowledge and their organization and management. Several of those are engaged with the NAIRR Pilot.

The full-scale implementation of NAIRR will fundamentally support the advancement of AI research in the U.S. We respectfully encourage the federal government to include NAIRR in its AI Action Plan, and to prioritize the launch of the competitive award process for the NAIRR Operating Entity (OE). As outlined in the NAIRR blueprint, the OE, will efficiently organize the access, training, and governance related to all resources in NAIRR, critically bridging gaps across the multi-sector AI landscape in support of world-leading AI research and real-world impact. The blueprint additionally envisions that resources be funded via a NAIRR program, in addition to all of the contributed resources across government, industry, philanthropy, and academia. For decades, university and federal research centers have served as a national asset by innovating at the frontiers of science and technology; this work has significantly contributed to the nation's current status as a world leader in AI. These centers are strategically positioned to continue to lead and enable innovations at the forefront of AI. **Sustaining and advancing the nation's AI dominance is time-sensitive and urgent in this moment of extreme global competition. Thus, we recommend the expeditious implementation of NAIRR through the federal government's AI Action Plan in a way that fully leverages the leadership that university research centers can provide.**